

VOLUME Appendix

Appendix: Reproduction Protocols

Environment, seeds, data dictionary, control register, scorecard

Ishak Khamzatovich Isaev

Independent researcher, Nalchik, Kabardino-Balkarian Republic

Appendix: Reproduction Protocols

Environment, seeds, data dictionary, control register, scorecard
Ishak Khamzatovich Isaev · Independent researcher, Nalchik, Kabardino-Balkarian Republic
Hilbert Polya Bridge · monograph series C1–C8 · том Appendix · версия 1.0.0 · October 10, 2026

Contents

Environment and determinism	2
Data dictionary	2
Register of positive and negative controls	2
The Hilbert–Pólya scorecard	3

Environment and determinism

All campaigns ran in one environment: Python 3.12, numpy 2.1, scipy 1.14 (eigvalsh, driver evr), mpmath 1.3 (dps 25–40 per criterion), matplotlib 3.9 (600 dpi figures). The master seed of the series is 96; the RNG streams are fixed: vortices — `default_rng(96 + 3500)`, disorder — `default_rng(96 + 3501)`, GUE nulls — `default_rng(9000 + k)`, Poisson nulls — `default_rng(7000 + k)`. Runtimes: C1 ~2 s, C2 ~30 min (two arithmetics), C3 ~3 min, C4 ~12 s, C5 ~5 s, C6 ~8 min (the 10^6 sieve), C7 ~20 min (L=96 dynamics), C8 ~1 min.

The bit-for-bit base control: the protocol reference gate reproduces the mean level spacing of the C3 CSV with deviation $1.45\text{e-}16$ — the kernel port (`engine/hamiltonian.py`) is equivalent to the original protocol up to summation order. All verdicts are committed in `data/`; the campaigns re-read them and recompute cheap invariants in seconds.

Data dictionary

The ζ zero file (2 001 052 heights, Odlyzko format) is not shipped for licensing reasons; its path is set by `ZFILE` in `campaigns/` and defaults to `verification/data/zeros6.txt` of the neighbouring repository. Every verdict of the series is committed, so no test requires the zero file; fresh campaigns with recomputation are optional.

Table 1. Verdict files

File	Content	Campaign
<code>c1_identity_verdict.json</code>	K1–K7: ladder, reality, zeros, Gram	C1
<code>c2_argument_verdict.json</code>	anchors 0/3/10/29/79/649, S(T), gaps, bridge	C2
<code>c3_gates_verdict.json</code> + <code>c3_gates_table.csv</code>	52 gates, nulls, PC1/PC2	C3
<code>c4_trace_verdict.json</code>	M0/M1/M2, PC0, trace-metric gap	C4
<code>c5_gates_ext_verdict.json</code> + <code>c5_gates_ext_table.csv</code>	35+2 extended gates	C5
<code>c6_primes_verdict.json</code>	co-location, kernel, ψ -ladder, codings	C6
<code>c7_dynamics_verdict.json</code>	trajectories $\langle x \rangle(t)$, transport $P(x \geq 90)$	C7
C8 ladder + spinor64	11-channel ladder, 64→4 reduction	C8

Register of positive and negative controls

The discipline of controls: a method is allowed a negative verdict only after demonstrating a positive result on an object where identity is known to hold. Every negative result of the series (H2, H5, M3) passed this procedure; every positive one (T1–T4, T6, T8) passed an independent recount in IVP01–IVP10.

Table 2. Controls of the series

Control	Must show	Value	Where
PC1 spinor64 twins	$r = 1$ on an isospectral orbit	$r = 1.000000000$, $D_{nn} = 2.7e-14$	C3
PC2 graph $\leftrightarrow$ file	$r = 1$ between Z_{id} zeros and the file	$r = 1.000000000000$	C3
PC0 zeros window (trace)	the method sees the Λ -comb	$r_{EF} = +0.9998$, residual 2.1%	C4
Decoys (random phases)	the null zone of correlations	0.026 ± 0.140 / $+0.050 \pm 0.153$	C3, C6
GUE null	universality without identity	$\langle r \rangle$ -plateau; $r_{EF} 0.43 \pm 0.71$	C3–C5
Surrogate null (blocks)	the width of the trace metric	0.00 ± 0.39	C4
Poisson null	clustered spectra	0.08 ± 0.80	C4

The Hilbert–Pólya scorecard

Table 3. The five questions of the programme

Item	Question	Verdict	Basis
H1	self-adjointness of the operator	YES	Hermitian by construction; real spectrum
H2	spectral identity level $\leftrightarrow$ zero	NO	9–10 orders (C3); ≈ 0.5 order (C4); $ r_{H7} \leq 0.043$ (C5)
H3	the GUE class	YES	$\langle r \rangle = 0.60$ at $N_v \geq 8$; all of C5 in class
H4	Dirac dynamics	YES	coherent packet, energy $1e-15$ (C7)
H5	primes in the operator	NO	ψ -ladder 0.972 for zeros, null zone for gates (C6)

The cycle formula after eight campaigns: the resonator is built and tuned (H1+H3+H4); the zeros are encoded, not generated (H2+H5); the prime weight is verified only for the zero operator (T6). The roadmap — in results/ROADMAP.md: stitching the statistical and coding branches, nonlocal trace constructions, a number-theoretic explanation of T9.